

Indian Institute of Technology Jammu
Linux System Administration and Virtualization (ID-2-04(MO))
Session 2025-26 Semester 02
Class Quiz-1

Student ID: _____ Student Name: _____

Instructions: The use of unfair means (UFM), including the Use of Mobile Devices and the Internet, is not allowed during the exam. No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. All questions carry equal marks. Max Marks: 10

Q1. Which of the following sequences is in the **correct order**?

- (A) Power On, Boot Loader starts, POST checks, BIOS/UEFI wakes up, OS loads subsystems
- (B) Power On, BIOS/UEFI wakes up, POST checks, Boot Loader starts, OS loads subsystems
- (C) Power On, POST checks, BIOS/UEFI wakes up, Boot Loader starts, OS loads subsystems
- (D) Power On, BIOS/UEFI wakes up, POST checks, OS loads subsystems, Boot Loader starts

Q2. In Linux, which of the following is **not suitable** for knowing file system details

- (A) fdisk -l
- (B) diskpart -lv
- (C) df -hT
- (D) lsblk

Q3. In Linux, which of the following is **NOT correct**?

- (A) A swap partition is mandatory in Linux
- (B) User input is redirected to global sink /dev/null
- (C) /dev contains device files, representing each hardware device as a file
- (D) root file system is attached to / to extend the overall file system hierarchy

Q4. In Vi editor, what is the correct way to **substitute all occurrences** of text "Average" with "Good"?

- (A) :%s/Average/Good
- (B) :%s/Average/Good/g
- (C) :%sub/Average/Good
- (D) :%sub/Average/Good/g

Q5. Which of the following commands is used to **change the permissions** of a file in bash?

- (A) chown
- (B) chmod
- (C) chgrp
- (D) chattr -p

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Q6. In Vi editor, in command mode, which one of the following will **redo the last undone** changes?

- (A) r
- (B) Ctrl + r
- (C) u
- (D) Ctrl + u

Q7. In man pages, how do you **search for a word** or string?

- (A) :find word
- (B) /word
- (C) ?word
- (D) *word

Q8. Which of the following is the **most recent stable LTS release** of Ubuntu (as of Jan 2026)?

- (A) 24.01
- (B) 24.04
- (C) 24.10
- (D) 24.06

Q9. Which of the following commands can help print all directories in the PATH variable on separate lines?

- (A) echo \$PATH | grep -r ':' '\n'
- (B) echo \$PATH | tr ':' '\n'
- (C) cat \$PATH | grep -r ':' '\n'
- (D) cat \$PATH | tr ':' '\n'

Q10. From bash history, which one of the following can help execute the immediate previous command?

- (A) !history
- (B) !!
- (C) Ctrl + r
- (D) !1

P2

Answers:

1. B	2. B	3. A, B	4. B	5. B
6. B	7. B	8. B	9. B	10. B

Student ID: _____ Student Name: _____

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Answer the questions in the space provided. All questions carry equal marks. Max Marks: 10

A. Write regular expressions (regex) for the following (1 to 5):

1. Match all lines that start with \$, followed by any single digit, followed by \$, followed by one or more non-whitespace characters

`\$[0-9]\$[^\s]+`

2. Match every possible IPv4 IP address (use metacharacters and groups)

`([0-9]{1,3}\.){3}[0-9]{1,3}`

3. Match every 9-character string (with letters, numbers, and symbols) that doesn't end in a "!" sign

`[^\s]{8}[^!]`

4. Match PAN, examples AABPS0971Z, BBCYJ9876A, PANNO1234T

`[A-Z]{5}[0-9]{4}[A-Z]`

5. Match IFSC, contains 4 characters for bank name in capital, followed by one zero, followed by six characters in capital or numbers.

`[A-Z]{4}0[A-Z0-9]{6}`

6. How can you search a noun or verb using grep and ignoring "ing" in it? Use grep.

`grep -v "ing" text` or `cat text | grep -v "ing"`

7. How can you search a log file for ERROR and print five lines before and five lines after the matching line?

`grep -C 5 "ERROR" log.txt`

8. Write a command to replace all Unix references with Linux in the vi editor

`:%s/Unix/Linux/g`

9. Write four commands discussed in the class on networking.

`ip, iwconfig, ifconfig, netstat, nmap, dig, nslookup, ping, traceroute, ssh, telnet, nc, curl, openssl, wireshark, tcpdump`

10. Write a command that prints columns 1 and 2 of /etc/passwd using the separator ":"

`awk -F ':' '{print $1,$2}' /etc/passwd`

or `cat /etc/passwd | cut -d ':' -f 1,2 | tr ':' ' '`

Indian Institute of Technology Jammu
Linux System Administration and Virtualization (ID-2-04(MO))
Session 2025-26 Semester 02

Mid-Semester Exam

Student ID: _____ Student Name: _____

Instructions: The use of unfair means (UFM), including the Use of Mobile Devices and the Internet, is not allowed during the exam. No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. Section A - each question carries 05 Marks each, Section B - each question carries 02 Marks each, and Section C - each question carries 1 Mark each. MM-50

Section A

Q1. Write a bash script to calculate the sum of natural numbers from 1 to n , where n is a command-line argument. If n is not supplied, return error code 255. If n is not a non-negative number, return error code 254.

```
#!/usr/bin/bash
```

```
sum=0
```

^{swap}
Q2. Write a bash script to calculate the sum of two numbers, *num1* and *num2*, without using a third variable, where both *num1* and *num2* are command-line arguments. If any of these two numbers are not supplied, return error code 255.

```
#!/usr/bin/bash
```

```
sum=0
```

Section B

Q3. Write down permissions in **numeric representation** from the partial output shown.

lrwxrwxr-x	<u>775</u>	drwxr-xr-x	<u>755</u>
-r-----	<u>400</u>	-rw-r--r--	<u>644</u>
prw-----	<u>600</u>	srw-rw-rw-	<u>666</u>

Q4. Everything is file in Linux. The output of `ls -lrt`, column 1 lists **file type**. Fill in the blanks:

Regular file:	<u>-</u>	Directory:	<u>d</u>
Symbolic links:	<u>l</u>	Character device:	<u>c</u>
Block device:	<u>b</u>	Socket:	<u>s</u>

Q5. In the bash session, how can we modify the **PATH** variable to search the home directory for executables first? Make no assumptions about home; command should run everywhere.

export PATH=~/:\$PATH w/ tilde-slash 😊

or export PATH="\$HOME:\$PATH"

Q6. How will you grant read **permission** to the group users and other users? Command please.

chmod go+r file

Q7. How can we find multiple users who belong to the **adm** group using a command?

cat /etc/group | grep "^adm"

^adm because group name is the first column and to filter out other reference of adm, if any 😊

Q8. Inside the **Vi editor**, in command mode, how will you remove all occurrences of control characters ^V

: %s/^V^V//g

↪ to be removed
↓
marker for control characters

Q9. How can standard output and standard error be **redirected** to file log.txt? Write command.

&> log.txt

or command > log.txt 2>&1

check this out

<command> &> log.txt
you wish to execute

Q10. How to output the command output of `lsb_release -a` to the terminal and a file called release.txt also?

lsb_release -a | tee release.txt

or

lsb_release -a && lsb_release -a > release.txt

Q11. Complete the if statement in this bash script to match the output from echo.

file1=\$1

file2=\$2

IF_STATEMENT_HERE

←

echo "\$file1" is newer than "\$file2"

fi

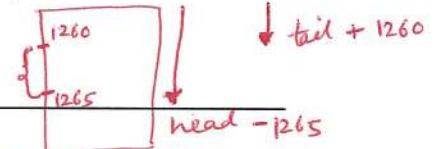
if [\$file1 -nt \$file2]; then

if [[...]]; then
works and permitted

Q12. rps wants to view lines from 1260 -to- 1265 in the file myfile.txt. Help him with the correct command. Hint: He recalls something from the head and tail.

head -1265 myfile.txt | tail +1260

or head -1265 myfile.txt | tail -6



option -n is permitted and works

by counting line difference

Q13. rps wants to share some files on the same network with msg using an HTTP server. He has Python installed. Help him with the correct command to start the server.

python3 -m http.server

Q14. rps has a dual-booted system with Windows and Linux (Ubuntu) installed. His system shows the incorrect time per the set timezone. He has booted into Ubuntu. What command should he use to synchronize the clock across operating systems? Mention the complete command.

timedatectl set-ntp true

or timedatectl set-local-rtc 1 --adjust-system-clock

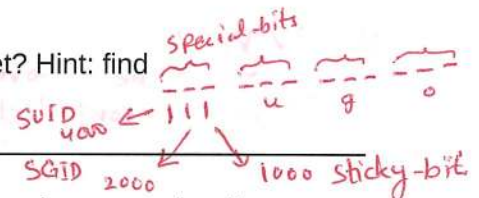
Q15. How will you schedule the bash script /home/rps/sync_attendance.sh to run every night at 11 pm, except on Sunday? Mention only the entry that goes in the cron job file.

00 23 * * 1,2,3,4,5,6 rps { cd && ./sync_attendance.sh }

↓
mon, tue, wed, thur, fri, sat

Q16. How can we query all executables that have the sticky bit set? Hint: find

find / -perm /1000



Q17. On Ubuntu, rps cannot recall if he set the root password. How do you confirm if no root password is set?

sudo -i

passwd -S

← check output, P = Password is set
NP = No password set
L = Password is locked

Section C

To answer the questions in Section C, fill in the following table:

1. <u>A</u>	2. <u>D</u>	3. <u>B</u>	4. <u>D</u>	5. <u>A</u>
6. <u>B</u>	7. <u>A</u>	8. <u>A</u>	9. <u>B</u>	10. <u>B</u>

Q1. Which of the following sequences is in the correct order?

- (A) Power On, BIOS/UEFI wakes up, POST checks, Boot Loader starts, OS loads subsystems
- (B) Power On, Boot Loader starts, POST checks, BIOS/UEFI wakes up, OS loads subsystems
- (C) Power On, POST checks, BIOS/UEFI wakes up, Boot Loader starts, OS loads subsystems
- (D) Power On, BIOS/UEFI wakes up, POST checks, OS loads subsystems, Boot Loader starts

Q2. In Linux, which of the following is NOT correct?

- (A) A swap partition is not mandatory in Linux
- (B) /dev contains device files, representing each hardware device as a file
- (C) The root file system is attached to / to extend the overall file system hierarchy
- (D) User input from standard input (stdin) is redirected to device /dev/null

Q3. Which of the following was a stable LTS release of Ubuntu?

- (A) 18.01
- (B) 20.04
- (C) 22.06
- (D) 24.10

Q4. From bash history, which one of the following can help execute the immediate previous command?

- (A) Page up arrow key
- (B) !!
- (C) !-1
- (D) All of the above

Q5. How to get a list of all signals on Linux, names, and numbers?

- (A) kill -l
- (B) kill -a
- (C) kill -9 -a
- (D) kill %1

Q6. A log file keeps updating and growing. How can we view its content while it is still getting updated?

- (A) head file
- (B) tail -f file
- (C) tail -z file
- (D) cat -t file

Q7. What is the numeric value of permissions for this file?

```
crw-rw-rw- 1 root root 1, 3 Feb 20 17:24 /dev/null
```

- (A) 666
- (B) 766
- (C) 777
- (D) 444

Q8. Which of the following is incorrect about the bash builtin cd?

- (A) cd takes to /home directory
- (B) cd ~ takes to the home directory
- (C) cd . . takes to parent directory
- (D) cd - takes to the previous directory

Q9. Execute the last command in the bash terminal that began with com

- (A) runlast com
- (B) !com
- (C) ^com
- (D) :&com

Q10. Which of the following commands locates the binary, source, and manual page files for a command?

- (A) plocate
- (B) whereis
- (C) which
- (D) whatis

*_*_*

Indian Institute of Technology Jammu
Linux System Administration and Virtualization ID-2-04(MO)

Session **2025-26** Semester **02**

[Max Duration: **3 hours**]

End-Semester Exam

[Max Marks: **100**]

Student ID(आई डी): _____ Student Name (नाम): _____

Instructions: The use of unfair means (UFM), including the Use of Mobile Devices and the Internet, is not allowed during the exam. No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. Section A - each question carries 10 Marks each, Section B - each question carries 03 Marks each, and Section C - each question carries 1 Mark each.

04

Section A

Q1. Write a bash script to **rename** WhatsApp*.jpeg files residing inside the directory /home/rps/Downloads/, and the files are:

'WhatsApp Image 2023-10-17 at 9.38.26 AM.jpeg'

'WhatsApp Image 2024-04-17 at 11.03.43 PM.jpeg'

'WhatsApp Image 2025-11-07 at 4.00.01 PM.jpeg'

'WhatsApp Image 2026-01-11 at 12.37.26 PM.jpeg'and so on.....

Assuming they are less than one thousand files, the script should rename them to sachin-001.jpeg, sachin-002.jpeg, sachin-003.jpeg and so on, to a **new** directory called /home/rps/Downloads/sachin because April 24 is his birthday and we are doing cleanup of Downloads directory. The script should return (exit) error code 255 with appropriate error message if there are no such files in the first place. Script should output the names of new files created successfully one file per line. Hint-1: WhatsApp*.jpeg and Hint-2: mv sourcefile targetfile

Answer:

```
#!/usr/bin/bash
```

```
# This script renames WhatsApp*.jpeg files to sachin*.jpeg files and numbers them properly.
```

```
# Short comments are nice, they tell us what is going on.
```

```
# Use meaningful variable names, and appropriate indentation for better readability.
```


Q2. Write a bash script to **create 30 new users**, namely user01, user02, user03, ..., user29, user30, and **set their respective passwords** as p@\$\$w0rd01, p@\$\$w0rd02, p@\$\$w0rd03, ..., p@\$\$w0rd29, p@\$\$w0rd30. We will share with them later by email. Also, **set the password to expire** so that users are prompted to change their password upon the first login. Script should output user created, and the password in separate lines followed by 1 empty line.

Hint-1: adduser and Hint-2: chage

```
#!/usr/bin/bash
```

```
# This script create users with initial login password; user needs to change password upon login
```

```
# Short comments are nice, they tell us what is going on.
```

```
# Use meaningful variable names, and appropriate indentation for better readability.
```


Q3. [] Write a bash script to **check the format** of a command line argument (input) **IP v4 address of the form W.X.Y.Z** using regular expressions. You are aware that in a IP v4 address, each octet W, X, Y, or Z can vary from 0 to 255, or each octet can be of 1, 2, or 3 digits. Script should return error code 255 if no argument to the script is supplied.

OR

Q3. [] Write a bash script to **check the format** of a command line argument (input) **text as a maximum of three syllable word name** using **regular expression**. Each word, if present, begins with a capital letter and there is exactly one space between these words. There is **no trailing space**. For example, valid names are:

"S Arun Kumar", "Sam Arun K", "Sam A Kumar", "Sam Arun Kumar", "S Arun", "Sam", "S Arun K", "S A K"

while "SArunKumar", "S arun kumar", "arun kumar", "S Arun Kumar", "SAK", "S Arun ", are not in acceptable format.

Note: In case, you **attempt both options** for evaluation, **you simply get ZERO marks** for this Q3. You are required to make a choice first.

Answer:

```
#!/usr/bin/bash
```

```
# This script __
```

```
# Short comments are nice, they tell us what is going on, which option I am attempting.
```

```
# Use meaningful variable names, and appropriate indentation for better readability.
```


Section B

Q4. Write down file permissions in **numeric representation** from the permissions shown.

lrwxr-xr--	<u>754</u>	drwxr-xr-x	<u>755</u>
--w-----	<u>200</u>	-rw-r--r--	<u>644</u>
prwx-----	<u>700</u>	srw-rw-rw-	<u>666</u>
lrwxr-----	<u>740</u>	-r--r--r--	<u>444</u>

Q5. In the bash session, how can we modify the **PATH** variable to search the home directory for executables first? Make no assumptions about home; command should run everywhere.

export PATH="\$HOME:\$PATH" ↓
~/ home-directory

or export PATH=~/:\$PATH

Q6. How will you grant and revoke read **permission** to the group users and other users? Command please.

Grant: chmod go+r file

Revoke: chmod go-r file

Q7. Convert and Merge:

a. How to convert an image hate.png to PDF love.pdf

convert hate.png love.pdf

b. How to merge three PDF files file1.pdf, file2.pdf and file3.pdf into one pdf file merged_file.pdf

pdftk file1.pdf file2.pdf file3.pdf cat output
merged-file.pdf

Q8. Why do system administrators prefer "su -" over a simple "su" command to substitute a user?

Q9. Mention two commands to list out both file systems and mount points.

df -hT

Q10. How can rps create 100 files in /tmp directory using touch? File names should be named as app001.txt, app002.txt, app003.txt,.....,app099.txt, app100.txt.

touch /tmp/app{001..100}.txt

Q11. Complete the if statement in this bash script to match the output from echo.

file1=\$1

file2=\$2

IF_STATEMENT_HERE

=

echo "\$file1" is older than "\$file2"

fi

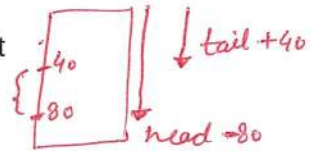
if [\$file1 -ot \$file2]; then

Q12. rps wants to view lines from 40 to 80 in the file time.txt. Help him with the correct command. Hint: He seems to have forgotten everything from head to toe (tail).

head -80 time.txt | tail +40

or head -80 time.txt | tail -40

by counting line difference



*option -n
is permitted
and niks*

Q13. One website URL <https://www.mysite.com/uploads/> shows all files uploaded by users or applicants. Anyone can delete them using HTTP DELETE [mysite.com/uploads/file](https://www.mysite.com/uploads/file) specific URLs. What is the issue here with the uploads directory, and how to fix it?

Issue uploads directory has write permission set for others

Fix remove/revoke write permission from others $\Rightarrow$ chmod o-w uploads

Q14. What is the difference between these two ways of executing a bash script?

\$ bash -v script.sh

\$ bash -x script.sh

bash -v echo or display the commands before running them

~~execute~~

<command>

execute command and show output

bash -x echo commands after processing them one-by-one

and expanding variables

execute command and show output

<command with variable expansion>

bash -v script.sh

bash -x script.sh

Q15. How will you schedule the bash script /home/rps/sync_attendance.sh to run every night at 11 pm, except on Sunday? Mention only the entry that goes in the cron job file.

00 23 * * 1,2,3,4,5,6 rps { cd && {bash sync_attendance.sh} }

↓
mon, ~~thu~~, wed, thu, fri, sat

Q16. Why is a non-root user able to execute a command owned by root?

Q17. User rps has a compressed file ~/Downloads/json.zip and he wants to know its compressed size, uncompressed size, compression ratio, and name of constituents inside the archive or compressed file. Help him with the correct command.

zcat -l ~/Downloads/json.zip

zipinfo ~/Downloads/json.zip

gzip -l

gunzip -l

Q18. Everything is file in Linux. The output of ls -lrt, column 1 lists file type. Fill in the blanks:

Regular file: - Directory: d

Symbolic links: l Character device: c

Block device: b Socket: s

Named pipe: p Executable file: -

Q12

tail +40 time.txt | head -n 41 -

head -40 time.txt | tail -n 41 -

Section C

To answer the questions in Section C, fill in the following table:

Tick or answer to these questions written outside this table shall NOT be considered.

1. <u>A</u>	2. <u>C,D</u>	3. <u>B</u>	4. <u>C</u>	5. <u>A</u>
6. <u>B</u>	7. <u>D</u>	8. <u>A</u>	9. <u>D</u>	10. <u>A</u>

Q1. Which of the following sequences is in the correct order?

- (A) Power On, BIOS/UEFI wakes up, POST checks, Boot Loader starts, OS loads subsystems
- (B) Power On, Boot Loader starts, POST checks, BIOS/UEFI wakes up, OS loads subsystems
- (C) Power On, POST checks, BIOS/UEFI wakes up, Boot Loader starts, OS loads subsystems
- (D) Power On, BIOS/UEFI wakes up, POST checks, OS loads subsystems, Boot Loader starts

Q2. In Linux, which of the following is NOT correct?

- (A) Everything is stored as file
- (B) Every process cannot be preempted
- (C) Every process is stored as a file in /proc
- (D) Every named pipe file is stored in /dev

Q3. Canonical has shipped the next stable LTS release of Ubuntu yesterday, April 23, 2026.

Which of the following is the stable LTS release that was released earlier?

- (A) 18.01
- (B) 20.04
- (C) 22.10
- (D) 24.12

Q4. On a Linux system, generally, which one of the following is most likely to have one or more socket files?

- (A) /dev
- (B) /proc
- (C) /run
- (D) /var

Q5. On a bash terminal, what is the output of command **yes**

- (A) continuous print of y
- (B) continuous print of yes
- (C) continuous print of . (dot characters)
- (D) continuous print of output of ping localhost

Q6. A log file keeps on updating and growing on Linux system. Which utility can help to monitor and keep log files contained on defined frequency (weekly, monthly, yearly)?

- (A) logview
- (B) logrotate
- (C) logpurge
- (D) logcheck

Q7. What is the numeric representation of permissions for the three stream files, stdin, stdout, stderr? Recall we discussed in the class `ls -lrt /dev/std*`

```
l<permissions> 1 root root 15 Apr  8 17:08 /dev/stdout -> /proc/self/fd/1
```

```
l<permissions> 1 root root 15 Apr  8 17:08 /dev/stdin -> /proc/self/fd/0
```

```
l<permissions> 1 root root 15 Apr  8 17:08 /dev/stderr -> /proc/self/fd/2
```

- (A) 444
- (B) 555
- (C) 666
- (D) 777

Q8. Which one of the following is incorrect about the bash builtin cd?

- (A) `cd` takes to /home directory
- (B) `cd ~` takes to the home directory
- (C) `cd .` takes to the parent directory
- (D) `cd -` takes to the previous directory

Q9. Given that the Linux Systems class of session 2025-26 semester 02 had a total of 40+ hours of classroom engagement hours, two in-class quizzes, one midsem, one project, one poster, one endsem, and 6 lab handouts, how many hours you have contributed to this course in total including class hours, practice and preparation hours?

- (A) 0 to 40 hours (just attended a few classes, and browsed two-three GitHub pages)
- (B) 40 to 80 hours (more than 70% of the classes and preparation for mid sem, end sem and submissions)
- (C) 80 to 112 hours (more than 80% of the classes and 2 to 4 hours per week average outside the class hours)
- (D) 112 to 140 hours or more (more than 80% of the classes and 4 to 6 hours per week outside the class hours)

One week has $24 \times 7 = 168$ hours. How many hours did I give?

Q10. Which one of the following best describes the number of doubts asked of all students (~30) registered in the Linux Systems class of session 2025-26 semester 02, by visiting the office room of the instructor or taking some time to meet in Pushkar or Mansar building outside designated class hours and minutes around them?

- (A) less than 5 students on less than 5 occasions in total
- (B) less than 10 students on less than 10 occasions in total
- (C) less than 15 students on less than 15 occasions in total
- (D) less than 20 students on less than 20 occasions in total

All the best for summer break 😊 learn, build, nurture something nice that you will be proud of.

*_*_*

shell scripting - modern syntax (())
over classical → [[]] —
←